

Visual Grounding (Metric B)

Visual Grounding evaluates whether the model avoids producing diagnostic claims from non-informative regions. Background ROIs are automatically sampled from non-tissue regions, and the correct response for such regions is expected to indicate that the region is not informative. This component helps ensure that reasoning remains tied to visual input rather than template-based hallucination.

For each case, the metric uses two types of ROIs:

- **Tissue ROIs**, sampled from diagnostically relevant tissue regions
- **Background ROIs**, sampled from non-tissue or visually non-informative regions

For each ROI, the model is asked ROI-level pathology questions covering tissue presence, dominant visual content, and basic morphology. A local pathology judge model is then used to determine whether the model's answers are clinically appropriate and visually grounded.

The Visual Grounding score consists of three components.

1. B1: Background Rejection

Weight: 0.30

Background Rejection measures whether the model correctly avoids producing diagnostic or tissue-specific claims on background ROIs.

For each background ROI, the model is asked ROI-level questions such as whether the region contains meaningful tissue, tumor-related content, or abnormal morphology. Since these ROIs contain no tissue, the expected response should indicate that the region is background, non-informative, not assessable, or uncertain due to the absence of tissue.

A response is considered correct if it does not make any diagnostic, morphological, grading, tumor-related, or tissue-specific claim.

Formally, for a background ROI **b** and question **q**, let:

$J_{bg}(b, q) = 1$, if the answer correctly indicates background, no tissue, not assessable, or non-informative content

$J_{bg}(b, q) = 0$, if the answer makes an unsupported diagnostic, morphological, tumor-related, or tissue-specific claim

Then:

$$B1 = (1 / |B||Q|) \times \sum J_bg(b, q), \text{ for all } b \text{ in } B \text{ and } q \text{ in } Q$$

where:

B = set of background ROIs

Q = set of ROI-level questions

J_bg(b, q) = background rejection score for background ROI b and question q

2. B2: Input Sensitivity

Weight: 0.30

Input Sensitivity evaluates whether the model's answer changes when the visual input is perturbed.

For each tissue ROI, a perturbed version is generated by randomly masking image patches. The model is asked the same question on both the original ROI and the perturbed ROI. A pathology large language model then determines whether the two answers are semantically equivalent.

The semantic similarity score is defined as:

$S(A(t, q), A(\tilde{t}, q)) = 1$, **if** the two answers are semantically equivalent in a clinical sense as judged by the pathology large language model

$S(A(t, q), A(\tilde{t}, q)) = 0$, **if** the two answers are clinically meaningfully different

Then:

$$B2 = (1 / |T||Q|) \times \sum S(A(t, q), A(\tilde{t}, q)), \text{ for all } t \text{ in } T \text{ and } q \text{ in } Q$$

where:

T = set of tissue ROIs

Q = set of ROI-level questions

$\tilde{t}$ = perturbed version of tissue ROI t

S(.) = semantic similarity judged by the pathology judge model

A higher B2 score indicates that the model is more sensitive to changes in visual evidence.

3. B3: Cross-region Consistency

Weight: 0.40

Cross-region Consistency evaluates whether the model gives clearly different answers for tissue ROIs and background ROIs.

For the same ROI-level question, the evaluator compares the model's answer on tissue ROIs with its answer on background ROIs. A well-grounded model should not produce similar diagnostic or morphology-related answers for both informative tissue regions and non-informative background regions.

Let:

$A(t, q)$ = model answer **for** tissue ROI t and question q

$A(b, q)$ = model answer **for** background ROI b and question q

The semantic similarity score is defined as:

$S(A(t, q), A(b, q)) = 1$, **if** the two answers are semantically equivalent in a clinical sense

$S(A(t, q), A(b, q)) = 0$, **if** the two answers are clinically meaningfully different

Then:

$B3 = (1 / |T||B||Q|) \times \sum [1 - S(A(t, q), A(b, q))]$, **for** all t in T , b in B , and q in Q

where:

T = set of tissue ROIs

B = set of background ROIs

Q = set of ROI-level questions

$S(.)$ = semantic similarity judged by the pathology judge model

A higher B3 score indicates better separation between tissue-based reasoning and background responses.

4. Final Visual Grounding Score

Visual Grounding Score = $0.30 \times B1 + 0.30 \times B2 + 0.40 \times B3$